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Overcoming the Limitations of CNN-Based AI Skincare Predictions: A ConvLSTM Approach for Enhanced Personalization

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Abstract

Personalized skincare is limited by static, single-image CNNs that ignore temporal change and often underperform across diverse skin tones and capture settings. This paper aims to evaluate whether a ConvLSTM multimodal approach can better model user-specific skin trajectories to enhance personalization and recommendation stability. Literature shows that temporal dynamics are under-modelled, calibration and performance degrade under variable capture conditions, subgroup reporting and fairness are inconsistent, and the clinical ground truth is often weak. The research methodology identifies stratified sampling, clinician annotation, and integration of environmental covariates to increase the accuracy of the system and enhance personalization. Evidence indicates that temporal modelling with multimodal signals and standardized capture improves personalization accuracy and recommendation stability. The proposed system envisions advancing equitable, clinically grounded skincare by modelling a user's changing skin to deliver personalized recommendations.

Index Terms:
CNN, ConvLSTM, Fairness, Multimodal inputs, Personalized skincare, Spatiotemporal modelling, Standardized capture.

1.0 Introduction

Personalised skincare remains constrained by static, single-image CNN systems that overlook how skin changes with environment, lifestyle, and physiology. In the previous proposal, this problem was analysed and a ConvLSTM-based approach that learns temporal skin trajectories for better personalization and accuracy, moving beyond one-off CNN predictions was proposed. It also stated the research aim to validate ConvLSTM for personalised skincare and motivated the shift to multimodal inputs that fuse images with diaries and contextual signals. In addition, the previous proposal outlined a four-lane system architecture which included, data ingestion, feature engineering, ConvLSTM forecasting, and decision-feedback. Features that have been analysed to deliver adaptive recommendations, and it highlighted key limits of snapshot CNNs, including bias, lighting sensitivity, and lack of feedback.

Building on that foundation, this paper provides a systematic literature review and a methodology that elaborate the technical and conceptual background of the proposed system. The review covers three areas: (i) snapshot CNN skincare systems and their limitations, (ii) temporal modelling with ConvLSTM, and (iii) multimodal personalization that fuses images with routine and context. It also examines similar multimodal systems to assess the validity of the approach. The methodology then specifies target users, sampling, standardized capture and normalization, clinician annotations for calibration, and integration of secondary environmental data to evaluate the system under realistic conditions. By using this approach, this review aims to show that enhanced

personalization and improved recommendation accuracy can be achieved.

2.0 Literature Review

2.1 Research domains

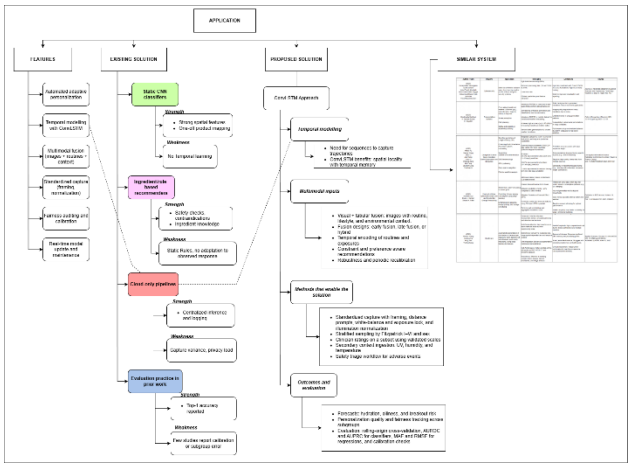

Figure 1: Literature Review Map

This literature review is structured according to the literature map in Appendix Figure 1. It first outlines the main limitations of existing static CNN-based skincare systems, including dataset bias, capture variance, and the lack of temporal learning. It then highlights how ConvLSTM and multimodal inputs are positioned as a trajectory-aware solution that links spatial features with routine and context signals. Next, comparable existing system in other domains are considered as benchmarks to show what current systems do well and where they fail. Finally, the map connects the proposed approach to its planned evaluation methods, including standardized capture, stratified sampling, clinician input, and temporal and fairness-focused performance metrics.

2.1.1 AI for skin assessment and recommendation

2.1.1.1 Snapshot CNN capabilities, ecosystem gaps, and capture variance

a) Snapshot CNN classifiers and static recommenders.

Early consumer and research systems typically rely on convolutional neural networks that process single images to infer skin type, detect concerns such as acne or redness, and then map those outputs to a fixed set of products or routine steps (Sharma et al., 2025; Yadav & Singh, 2025; Lee et al., 2024). These workflows capitalize on the strength of CNNs in spatial feature extraction, especially for texture, pore visibility, and erythema cues. However, because each image is treated independently, the model neither encodes temporal context nor learns how skin responds after a routine change. This snapshot framing constrains personalization: two users with similar appearances today may be on very different trajectories, yet a single-image predictor cannot differentiate them. In routine use, the result is advice that is competent for static identification but often repetitive, generic, and slow to reflect real change.

b) Dataset bias, representation, and fairness.

A central limitation in static snapshot systems concerns the composition of training data. When darker skin tones, atypical presentations, or rare concerns are underrepresented, model behaviour can be unstable or biased for those groups (Siddiqui et al., 2025; Rajegowda et al., 2025). Because many pipelines also lack structured follow-up labels, they do not learn from the consequences of their recommendations. This weak feedback loop allows errors to persist and can amplify disparities. Practically, bias may appear as underestimation of inflammation on deeper tones or the over-recommendation of

actives that elevate irritation risk in sensitive users. Without targeted auditing and subgroup reporting, these gaps remain hidden behind aggregate metrics.

c) Capture variance and device heterogeneity.
Performance in the wild is further affected by illumination, pose, distance, and camera processing. Front cameras vary in optics, dynamic range, and white balance tuning, while device software introduces sharpening, smoothing, or tone mapping that alters apparent texture and colour. Snapshot CNNs are sensitive to these factors, which can produce day-to-day volatility and hinder cross-device generalization unless strong normalization and calibration strategies are used (Sharma et al., 2025; Lee et al., 2024). Even well-trained models may shift their severity estimates when the user moves from daylight to indoor lighting or switches phones mid-study.

2.1.1.2 Knowledge grounding, reporting practice, and motivation for temporal modelling

d) Knowledge grounding and rule layers.
Some systems augment CNN outputs with ingredient knowledge or rule-based contraindication checks. This increases safety and interpretability, yet the rules often remain static. Without temporal evidence or adherence tracking, the rules cannot adapt to the user's observed responses over weeks or seasons. Consequently, recommendations may remain conservative but generic, missing opportunities for individualized escalation or de-escalation informed by observed outcomes (Lee et al., 2024).

e) Evaluation gaps and reporting practice.
Published results frequently emphasize top 1 accuracy or similar single-image metrics. Few reports include probability calibration, subgroup error across Fitzpatrick types, or robustness under lighting and device shift, even though these properties determine whether advice is usable for diverse users in real settings (Siddiqui et al., 2025; Rajegowda et al., 2025). The absence of longitudinal metrics also obscures whether a system stabilizes or degrades as conditions change, thereby limiting conclusions about reliability over time.

The CNNs that use snapshot image capture has been effective for static recognition but does not capture how skin changes or how users respond to routines. The limitations above motivate temporal modelling and richer input signals that can support adaptive, fairness-aware recommendation.

2.1.2 Temporal modelling in vision

2.1.2.1 Sequence learning needs and ConvLSTM formulation

a) Sequence learning needs in skincare.
Skin evolves along trajectories influenced by product exposure, adherence, sleep, stress, seasons, and environment. Independent-frame classifiers cannot represent lagged effects, cumulative exposure, or cyclical patterns. As a result, they struggle to forecast outcomes, to attribute improvements to specific behaviours, or to differentiate temporary flares from persistent trends.

b) ConvLSTM formulation and benefits.
Convolutional LSTM replaces the fully connected gates of a standard LSTM with convolutions, allowing the recurrent cell to operate on spatial feature maps while learning temporal dependencies. This architecture preserves locality, so patterns like gradual reduction in T-zone oiliness or slow

improvement in barrier function can be modelled as sequences rather than isolated snapshots. Empirical results from spatiotemporal forecasting tasks such as Earth orientation prediction and residential load forecasting demonstrate improved accuracy and stability when temporal dynamics are essential, indicating suitability for skincare sequences where both space and time carry signal (Yu et al., 2025; Oza et al., 2025).

2.1.2.2 Training, explainability, missingness, and summary rationale

c) Training strategies for stability and horizon control.

ConvLSTM can be trained with sequence-to-one heads for next-week risk or sequence-to-sequence heads for daily forecasts. Curriculum schedules that gradually lengthen sequences help mitigate vanishing gradients and improve convergence on longer horizons. Regularization can include dropout within the recurrent stack and mix-up or augmentation at the frame level to simulate capture variation. These strategies produce models that not only fit the training data but also retain predictive sharpness as the horizon extends.

d) Explainability across time.

Explainability research has shown that static class activation mappings tend to highlight only the most discriminative region within a single frame, leading to incomplete pseudo-labels and weak object coverage (Aysel et al., 2025). In a temporal setting, this can manifest as flickering saliency and unstable rationales. Spatiotemporal attention and sequence-aware attribution can stabilize explanations, align rationales to persistent regions, and better reflect the evolution of signs across days. This is important for user trust and for clinician review of model outputs.

e) Handling irregular and missing sequences.

Real-world diaries are imperfect. Temporal models must handle irregular sampling and missing entries without over-interpreting gaps. Practical options include last-observation carry-forward with missingness flags, learned embeddings that encode “missing” as a state rather than zero, and simple adherence priors that prevent optimistic bias when users skip logging. Together, these techniques preserve temporal structure while acknowledging variability in user behaviour.

ConvLSTM provides a principled core for trajectory-aware prediction because it couples spatial feature learning with temporal memory. When combined with careful training, evaluation that prevents leakage, and sequence-level explainability, it addresses key failure modes of snapshot systems and brings modelling closer to how skin changes.

2.1.3 Multimodal inputs for personalization

2.1.3.1 Visual and tabular fusion and architecture designs

a) Visual plus tabular fusion.

Images are indispensable for spatial signs such as pores, shine, erythema, and texture. Yet many drivers of change are non-visual: ingredient concentrations and interactions, step-level adherence, and environmental context including UV index, humidity, and temperature. Studies that incorporate ingredient-aware and condition-aware signals with images report better alignment of recommendations to user needs, particularly when routines are complex or lighting varies between captures (Lee et al., 2024; Hash et al., 2025). Fusing these inputs enlarges the information set from “what the skin looks like today” to “what the user did, under

which conditions, and how the skin subsequently responded."

b) Early, late, and hybrid fusion designs.

Fusion architecture influences both accuracy and interpretability. In early fusion, per-day tabular embeddings are concatenated with CNN feature maps before temporal modelling, enabling the ConvLSTM to learn cross-modal interactions such as "retinoid at night plus low humidity increases irritation risk the next morning." Late fusion encodes visual and tabular sequences separately, then aggregates with attention or a gating mechanism to produce calibrated risk; this can enhance modularity and robustness when one modality is partially missing. Hybrid designs combine both, using early fusion for established interactions and late fusion to capture residual dependencies.

2.1.3.2 Temporal encoding, recommendations, robustness, and summary

c) Temporal encoding of routines and exposures.

Temporal modelling of non-visual factors is essential, since many exposures act cumulatively or with delay. Encoding sequences of routine steps, ingredient profiles, and context allows the model to separate persistent user traits from transient fluctuations and to forecast hydration, sebum change, or breakout risk under real-world rhythms (Hash et al., 2025). The result is advice that adapts to both who the user is and what the user does.

d) Constraint-aware and preference-aware recommendations.

Multimodal inputs also allow conditioning on user goals, sensitivities, budgets, and product constraints. Models can translate risk forecasts into actionable adjustments such as frequency changes or substitutions that respect user preferences. At the same time, fairness can be monitored by tracking subgroup error and calibration across Fitzpatrick types to ensure that benefits accrue equitably (Lee et al., 2024).

e) Robustness, generalization, and practical concerns.

Because diaries and sensors are noisy, robustness requires both data-level and model-level strategies. Augmentations that mimic device and lighting shift help reduce overfitting to narrow capture regimes, while exposure to diverse routines and climates improves generalization. Periodic recalibration maintains probability accuracy as seasons, habits, or product portfolios change. Handling missingness explicitly and using simple adherence priors keep the model conservative when information is incomplete.

Multimodal, temporally encoded inputs move the system from static, one-off classification to dynamic, context-aware personalization. By combining images with routine, lifestyle, and environmental signals, and by selecting appropriate fusion strategies, the model can forecast near-term outcomes and tailor recommendations more faithfully to how skin behaves in everyday life (Lee et al., 2024; Hash et al., 2025).

2.2 Similar systems

Author (Year)	Industry	Application	Strengths	Limitations	Source
(2025) Karagozoglu Karagozoglu Azeb Aynaw Mohd Tauhik Abdullah Mohd Izzam Ishak Tinggal Muhammad Daniel Hafiz Abdullah Yousif Raad Muhsen	Cybersecurity	Detection of fileless malware using deep learning models (ConvLSTM) in endpoint security systems	High detection accuracy (98%) Reduced processing time (10s vs 22s for DL STM) Lower loss rate Efficient spatial temporal feature extraction	Input size constraints with Conv2 STM10 Accuracy fluctuations / high loss in early training Need for improved visualization and reporting	Journal of Advanced Research in Applied Sciences and Engineering Technology. Volume 54, Issue 4, Pages 136-157
(2025) Kandapaty Madhavi K. Naveen Kumar G. Gayathri	Transportation Systems	Forecasting transitional weather conditions (e.g., sunny to rainy, foggy to sunny) to improve: Scene perception Path planning Safety and reliability in autonomous driving	Introduces TWFNet, a novel end to end spatio temporal forecasting framework Joint training of feature generation and classification improves accuracy Introduces AVIFD6, a realistic dataset for transitional weather forecasting Achieves high accuracy (up to 97.49%) in forecasting transitional weather status Demonstrates generalization to unseen weather transitions	Static backgrounds in generated transitions reduce real world applicability Image quality degradation in rainy transitions due to noise. Limited diversity in unusual weather patterns. Vulnerability to adversarial perturbations in noisy conditions Performance drop on simulated datasets like SHIFT compared to real-world AVIFD6	Pattern Recognition (Elsevier). DOI: 10.1016/j.patrec.2025.11.2154
(2025) Mengke Yu Xiaoya Wang Zhao Li Jian Wang Lihua Li Weiping Jiang Kai Feng Jingwen Long *Yuanhao Li	Geospatial Science, Space Geodesy, Earth Observation and Navigation	Real-time prediction of Length of Day (LOD) Enhancing Earth Orientation Parameters (EOP) forecasting Supporting satellite orbit determination (GNSS meteorology) RTK positioning Deep space navigation Extreme weather analysis	Integrates ConvLSTM (CNN + LSTM) for both short- and long term prediction capabilities Uses multi source datasets: IERS EOP C04, GNSS RRT LOD, and EAM (Effective Angular Momentum) Achieves: 56.70% improvement in ultra-short-term (1–10 days) prediction 10.07% improvement in short-term (10–30 days) prediction 11.90% improvement in medium- to long-term (60–300 days) prediction Addresses latency issues in traditional LOD data sources	Prediction accuracy varies with base sequence length Some deviations observed during specific cycles (e.g., ENSO influence) Requires high-quality, timely data from multiple sources Complexity in model tuning and tuning (e.g., learning rate schedules, sequence length optimization)	Geo-spatial Information Science. Published by Informa UK Limited (Taylor & Francis Group). DOI: 10.1080/10095020.2025.2471630
(2025) Abhishek Oza Bhuvan J. Ranger Dhaval K. Patel	Energy & Utilities, specifically Smart Grid and Residential Energy Forecasting	Residential Load Forecasting for smart grids Predicting 24-hour energy consumption using deep learning Enhancing grid planning, energy pricing, and storage scheduling	Extends forecast horizon to 24 hours Reduces prediction error by ~47% compared to other models Mitigates Persistence Forecast Effect (PFE) Combines spatial and temporal features using CNN and LSTM in parallel Performs well on individual and aggregated household loads	Trained on 2013 data, which may not reflect modern consumption patterns (e.g., EV charging) May not generalize well to atypical households Does not incorporate external factors like weather Requires annual retraining for optimal performance Model complexity may hinder scalability for large commercial buildings	Published in IEEE Access, Volume 13, 2025 DOI: 10.1109/ACCESS.2025.3528072
(2025) Weibin Huang Jiacong Cai Yang Yan Yuhong Kang	Healthcare	Automated segmentation of skin lesions in dermoscopic images for early detection of skin diseases, particularly melanoma, using deep learning techniques	Enhanced U-Net Architecture: Incorporates dense convolutional blocks and attention mechanisms. Multi Scale Attention: Improves focus on lesion-relevant features while suppressing noise. Bidirectional ConvLSTM: Captures long-range spatial dependencies and temporal context. GAN Integration: Enhances segmentation robustness and generalization. High Performance: Achieved state-of-the-art results on PH2, ISIC2017, and ISIC2018 datasets. Robustness: Effective in handling complex lesion shapes, blurred boundaries, and image artifacts.	Model Complexity: High computational cost due to dense architecture and multiple modules. Resource Intensive: Requires significant GPU memory and processing power. Small Lesion Performance: Struggles with extremely small or low-contrast lesions. Clinical Deployment: Needs further optimization for real-time or resource-constrained environments.	Applied Sciences, Volume 15, Article 8588 DOI: 10.3390/app15158588 Published by MDPI, June 10, 2025

Table 1: Summary of Similar System's using ConvLSTM in other industries.

This study draws on ConvLSTM applications in several spatiotemporal domains to justify a trajectory-aware skincare system. In space geodesy, Yu et al. (2025) integrate effective angular momentum and rapid Length-of-Day series within a ConvLSTM to improve near real-time LOD prediction across multiple horizons. Their work shows that pairing heterogeneous time series inside a unified spatiotemporal model can overcome latency in traditional sources, while revealing that sequence length and data quality strongly influence performance. In residential energy forecasting, Fusion ConvLSTM-Net combines parallel spatial and temporal encoders to extend the load forecast horizon and reduce error compared with CNN–LSTM hybrids, particularly by mitigating persistence effects where models simply repeat the last observed value (Oza et al., 2025). Together, these systems demonstrate that ConvLSTM can fuse multimodal context, control forecast horizon, and remain effective when local patterns evolve over time, which aligns closely with the needs of skincare trajectories driven by products, lifestyle, and environment.

Other fields provide complementary evidence without being examined in equal depth. In cybersecurity, ConvLSTM has been used for fileless malware detection with high accuracy and reduced inference time, showing that convolutional gating can capture compact local–temporal patterns under deployment constraints (Kareegalan et al., 2025). Transportation studies such as TWFNet treat weather transitions as spatiotemporal sequences, while medical imaging models like MA-DenseUNet with recurrent units show that temporal attention can stabilize feature focus across related frames (Huang et al., 2025; Madhavi et al., 2025). Across these domains, common lessons emerge: ConvLSTM improves forecasting when spatial structures change over time, but its benefits depend on careful handling of capture variance, dataset quality, calibration, and distribution shift. The proposed skincare framework therefore adopts ConvLSTM for sequence learning, fuses multimodal inputs, and emphasizes horizon tuning, augmentation, and efficiency so that temporal modelling translates into practical, reliable recommendations.

2.3 Proposed ConvLSTM Approach

Among the existing solution categories, cloud-only pipelines are the closest to enabling temporal modelling as they typically centralise and store sequential user logs, even though the models themselves remain static. However, none of the current implementations in skincare employ true temporal learning, and no system in prior work uses recurrent or ConvLSTM-based architectures. This highlights a significant research gap that the proposed system addresses.

The proposed system introduces a ConvLSTM-based skincare recommendation system that responds to the limitations of current CNN-driven tools. Existing skincare applications such as Smart Dermatology classify single images without considering how skin changes over days or weeks, so their recommendations remain static and weakly personalized (Sharma et al., 2025; Yadav & Singh, 2025). At the same time, evidence from other fields shows that ConvLSTM models handle spatial and temporal structure more effectively than conventional networks, with strong detection accuracy in cybersecurity, transportation, energy, and geoscience forecasting tasks (Kareegalan et al., 2025; Oza et al., 2025; Yu et al., 2025). Building on these findings, the proposed system treats skincare as a time-series prediction problem and aims to forecast skin trajectories rather than only labelling one snapshot.

Temporal modelling is central to the design. Instead of analysing a single selfie, the system ingests sequences of daily images, skincare logs, and environmental context such as weather and UV exposure. This allows the ConvLSTM layers to learn how redness, oiliness, or breakouts evolve in relation to routines and lifestyle patterns, like how ConvLSTM models track fileless malware behaviour over time or predict shifts in climate variables (Kareegalan et al., 2025; Yu et al., 2025). In contrast with CNN-only architectures that ignore user history and feedback (Hash et al., 2025; Yadav & Singh, 2025), the proposed system uses these sequences to adapt future recommendations as new data arrives.

Multimodal fusion further strengthens personalization. Image features from facial regions are combined with tabular signals such as product ingredients, usage frequency, sleep, stress, and exposure to sun or pollution. Studies in cosmetics and extended reality skincare already show that ingredient profiles and routine context influence perceived suitability of products (Lee et al., 2024; Rajegowda et al., 2025). ConvLSTM-based fusion models in energy and transport also report gains when visual or grid data are fused with time-stamped covariates (Oza et al., 2025). Inspired by these systems, the proposed architecture encodes both visual and non-visual inputs along a shared time axis so that recommendations respect user constraints such as allergies, lifestyle, and preferred product formats.

To make these forecasts reliable in practice, the system incorporates methods that standardize capture, support fairness, and enable safe deployment. Image collection will follow simple framing and lighting rules to reduce noise that previously harmed CNN accuracy on low-quality inputs (Yadav & Singh, 2025). Sampling and evaluation will be stratified by skin tone and sex to reduce the bias that has been reported for darker skin types in recent AI skincare work (Siddiqui et al., 2025). Finally, model quality will be tracked using metrics that mirror those used in other ConvLSTM applications, such as AUROC and AUPRC for classification and MAE or RMSE for continuous forecasts (Kareegalan et al., 2025; Oza et al., 2025; Yu et al., 2025). Together, these design choices move from one-off, opaque CNN predictions toward a temporally aware, multimodal system that can be updated, audited, and tuned in real time for diverse skincare users.

3.0 Methodology

3.1 Target Users

3.1.1 Direct users

Adults 18–55 who regularly use facial skincare and are willing to log routines and capture daily selfies using a smartphone. Direct users are adults aged 18–55 who use facial skincare and can complete daily logs and standardized selfies. To preserve fairness, we stratify by Fitzpatrick I–VI, sex, and age bands (18–25, 26–35, 36–45, 46–55) with balanced quotas. These users will benefit directly through personalized forecasts of hydration and oiliness, breakout-risk predictions for the next week, and adaptive routine recommendations. Guidance that stabilizes capture conditions, so longitudinal changes are meaningful will be ensured.

3.1.2 Indirect users

Dermatology professionals and allied health or beauty advisors who do not directly use the app but benefit from more consistent longitudinal evidence about a user's skin state. The study's standardized capture and normalization improve comparability across visits, which can support professional review and safety triage policies (Hernández et al., 2021; Shan et al., 2003).

3.2 Sampling Method

A stratified sampling approach will support fairness and subgroup evaluation. Participants are grouped by Fitzpatrick skin type I–VI and by sex, with quotas per group, in response to known performance gaps on darker tones and under-represented populations in dermatology AI (Daneshjou et al., 2022; Groh et al., 2021; Alipour et al., 2024).

Model performance will be reported both overall and by subgroup using AUROC and AUPRC for classification, MAE and RMSE for regression, and ECE for calibration. Subgroups include Fitzpatrick type, sex, and age bands. Each stratum will have a minimum size of 25 cases for training and 15 for evaluation; strata that do not meet this are oversampled or excluded from subgroup claims. Model drift will be monitored with quarterly recalibration checks.

3.3 Data Collection Method

The study uses a mixed data collection strategy that combines self-report diaries, standardized smartphone images, clinician ratings, and contextual data. This supports both model building and evaluation of user experience in a structured way (Saunders et al., 2023).

3.3.1 Daily in-app diary and imaging

Participants complete a brief daily diary in the app. Entries record products used, ingredient profiles, adherence, sleep, perceived stress, and optional menstrual phase and hydration. Each diary entry is paired with a standardized selfie. The app applies simple checks for blur, exposure, pose, and illumination, following current face-image quality and tele dermatology guidance (ISO/IEC, 2025; NIST, 2025; British Association of Dermatologists, 2024). Evidence linking sleep and stress with acne supports logging these variables (Schrom et al., 2019). Refer to appendix figure 2 for in app diary structure.

3.3.2 Weekly clinician ratings

Once a week, a blinded clinician rates acne severity using validated global grading scales such as IGA or GAGS. These scores act as ground truth for training and evaluating the model (Alsulaimani et al., 2020; Bae, 2024). Refer to appendix figure 5 for the rating structure.

3.3.3 Secondary context, safety, and governance

Daily UV index, temperature, and humidity are retrieved from public APIs and aligned with diary entries. Acne exposome research shows that solar radiation and climate influence severity, so these variables are included as contextual covariates (Dréno et al., 2018; Piquero-Casals et al., 2023). Participants can flag adverse reactions in the app, which pauses data capture and triggers clinician review, in line with safety expectations for non-interventional research (EMA, 2017; NIAMS/NIH, 2024). Participation is voluntary with informed consent, de-identified image storage, encryption in transit and at rest, role-based access, and a default retention period of 12 months. (Refer to appendix figure 3 and 4 for the checklist)

4.0 Conclusion

ConvLSTM-based multimodal skincare system is proposed to fix the limitations of static CNNs by learning time-varying skin. The system fuses standardized images with routine, lifestyle, and environmental data to improve personalization and accuracy. It treats evaluation, fairness, and governance as core design goals. Broader cohorts and clinician annotation help align decisions with real user needs and clinical practice. The results outline a blueprint for inclusive, trajectory-aware decision support in skincare. In the future, a working prototype will be built, and prospective trials will be conducted. Expansion of recruitment and labels, application of fairness-first training, and scheduling of regular recalibration. The robustness can be improved with domain adaptation and self-supervision. Uncertainty will be shown using conservative thresholds and routine clinician review. Privacy will be protected with federated or on-device learning. These steps move the design toward a safer system that delivers stronger personalization and steadier, context-aware recommendations.

Word Count: 3623

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6.0 Appendix

Figure 1

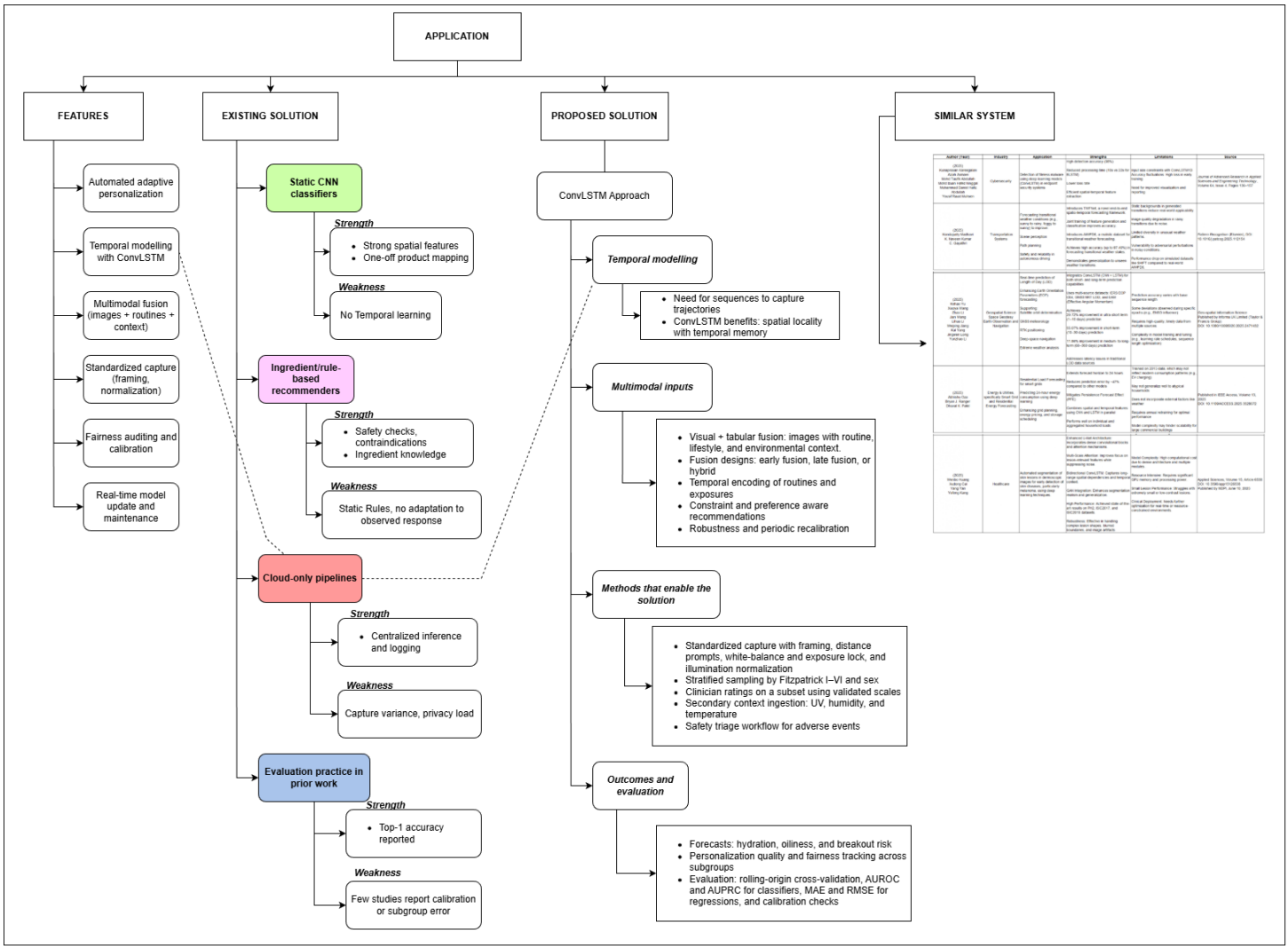
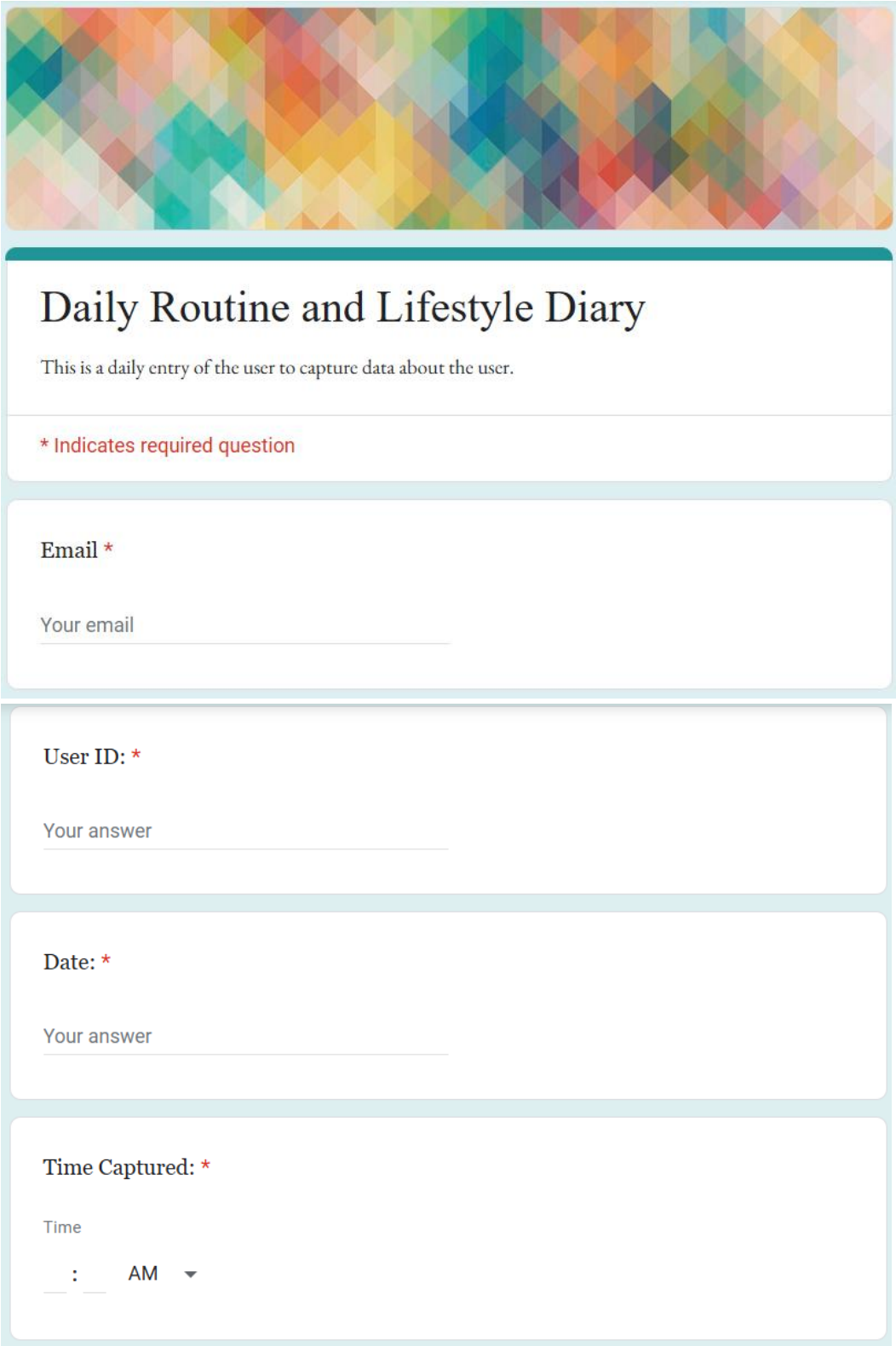


Table 1

Author (Year)	Industry	Application	Strengths	Limitations	Source
(2025) Kunaprasan Kareegalan Aziah Asmawi Mohd Taufik Abdullah Mohd Izuan Hafez Ninggal Muhammad Daniel Hafiz Abdullah Yousif Raad Muhsen	Cybersecurity	Detection of fileless malware using deep learning models (ConvLSTM) in endpoint security systems	High detection accuracy (98%) Reduced processing time (10s vs 22s for BLSTM) Lower loss rate Efficient spatial-temporal feature extraction	Input size constraints with ConvLSTM1D Accuracy fluctuations High loss in early training Need for improved visualization and reporting	*Journal of Advanced Research in Applied Sciences and Engineering Technology*, Volume 64, Issue 4, Pages 136–157
(2025) Kondapally Madhavi K. Naveen Kumar C. Gayathri	Transportation Systems	Forecasting transitional weather conditions (e.g., sunny to rainy, foggy to sunny) to improve: Scene perception Path planning Safety and reliability in autonomous driving	Introduces TWFNet, a novel end-to-end spatio-temporal forecasting framework. Joint training of feature generation and classification improves accuracy. Introduces AIWFD6, a realistic dataset for transitional weather forecasting. Achieves high accuracy (up to 97.46%) in forecasting transitional weather states. Demonstrates generalization to unseen weather transitions.	Static backgrounds in generated transitions reduce real-world applicability. Image quality degradation in rainy transitions due to noise. Limited diversity in unusual weather patterns. Vulnerability to adversarial perturbations in noisy conditions. Performance drop on simulated datasets like SHIFT compared to real-world AIWFD6.	*Pattern Recognition* (Elsevier), DOI: 10.1016/j.patcog.2025.112154
(2025) Kehao Yu Xiaoya Wang Zhao Li Jian Wang Lihua Li Weiping Jiang Kai Yang Jingwen Long Yunzhao Li	Geospatial Science Space Geodesy Earth Observation and Navigation	Real-time prediction of Length of Day (LOD) Enhancing Earth Orientation Parameters (EOP) forecasting Supporting: Satellite orbit determination GNSS meteorology RTK positioning Deep-space navigation Extreme weather analysis	Integrates ConvLSTM (CNN + LSTM) for both short- and long-term prediction capabilities Uses multi-source datasets: IERS EOP C04, GNSS NRT LOD, and EAM (Effective Angular Momentum) Achieves: 29.72% improvement in ultra-short-term (1–10 days) prediction 55.07% improvement in short-term (10–30 days) prediction 11.86% improvement in medium- to long-term (60–360 days) prediction Addresses latency issues in traditional LOD data sources	Prediction accuracy varies with base sequence length Some deviations observed during specific epochs (e.g., ENSO influence) Requires high-quality, timely data from multiple sources Complexity in model training and tuning (e.g., learning rate schedules, sequence length optimization)	Geo-spatial Information Science Published by Informa UK Limited (Taylor & Francis Group) DOI: 10.1080/10095020.2025.2471432
(2025) Abhishu Oza Bryan J. Ranger Dhaval K. Patel	Energy & Utilities, specifically Smart Grid and Residential Energy Forecasting	Residential Load Forecasting for smart grids Predicting 24-hour energy consumption using deep learning Enhancing grid planning, energy pricing, and storage scheduling	Extends forecast horizon to 24 hours Reduces prediction error by ~47% compared to other models Mitigates Persistence Forecast Effect (PFE) Combines spatial and temporal features using CNN and LSTM in parallel Performs well on individual and aggregated household loads	Trained on 2013 data, which may not reflect modern consumption patterns (e.g., EV charging) May not generalize well to atypical households Does not incorporate external factors like weather Requires annual retraining for optimal performance Model complexity may hinder scalability for large commercial buildings	Published in IEEE Access, Volume 13, 2025 DOI: 10.1109/ACCESS.2025.3528072
(2025) Wenbo Huang Xudong Cai Yang Yan Yufeng Kang	Healthcare	Automated segmentation of skin lesions in dermoscopic images for early detection of skin diseases, particularly melanoma, using deep learning techniques.	Enhanced U-Net Architecture: Incorporates dense convolutional blocks and attention mechanisms. Multi-Scale Attention: Improves focus on lesion-relevant features while suppressing noise. Bidirectional ConvLSTM: Captures long-range spatial dependencies and temporal context. GAN Integration: Enhances segmentation realism and generalization. High Performance: Achieved state-of-the-art results on PH2, ISIC2017, and ISIC2018 datasets. Robustness: Effective in handling complex lesion shapes, blurred boundaries, and image artifacts	Model Complexity: High computational cost due to dense architecture and multiple modules. Resource Intensive: Requires significant GPU memory and processing power. Small Lesion Performance: Struggles with extremely small or low-contrast lesions. Clinical Deployment: Needs further optimization for real-time or resource-constrained environments.	Applied Sciences, Volume 15, Article 6538 DOI: 10.3390/app15126538 Published by MDPI, June 10, 2025

Figure 2


Daily Routine and Lifestyle Diary
This is a daily entry of the user to capture data about the user.
* Indicates required question
Email *
Your email
User ID: *
Your answer
Date: *
Your answer
Time Captured: *
Time
: AM

Product used today in order: *
eg:
1. Cleanser [name + key ingredients]
2. Toner/Essence [name + key ingredients]
3. Moisturizer [name + key ingredients]
4. Sunscreen [name + key ingredients]

Your answer

Adherence flags: Did you skip a step? *

- Yes
- No

If Yes, which step and why? If No, then say N/A *

Your answer

Perceived Skin today: Hydration *

	1	2	3	4	5	
Dry	○	○	○	○	○	Hydrated

Perceived Skin today: Oiliness *

	1	2	3	4	5	
No oil	○	○	○	○	○	Very oily

Perceived Skin today: Irritation *

	1	2	3	4	5	
No irritation	○	○	○	○	○	Very irritated (Red inflammation)

Perceived Skin today: Breakout Count *

	1	2	3	4	5	6	7	8	9	10	
None	○	○	○	○	○	○	○	○	○	○	Severe Breakout

Hours Slept (last night) *

Choose

How stressed are you feeling? *

	1	2	3	4	5	6	7	8	9	10	
No stress	○	○	○	○	○	○	○	○	○	○	Very stressed

Menstrual phase *

Choose

How many ml of water did you drink? *

Your answer

Photo ID's: (Front, Left, Right [auto-assigned by app]) *

Your answer

Comments

Your answer

Figure 3

Standardized smartphone capture

User capture checklist

- ☐ Clean camera lens with microfiber cloth
- ☐ Remove heavy makeup and strong tint lip products
- ☐ Stand 45–60 cm from camera; face centered in guide
- ☐ Lock exposure and white balance on the face
- ☐ Use diffuse light source in front of face; avoid backlight
- ☐ Capture three photos: front, left 45°, right 45°
- ☐ Keep neutral expression and closed mouth
- ☐ Confirm auto-selected "best of three" image

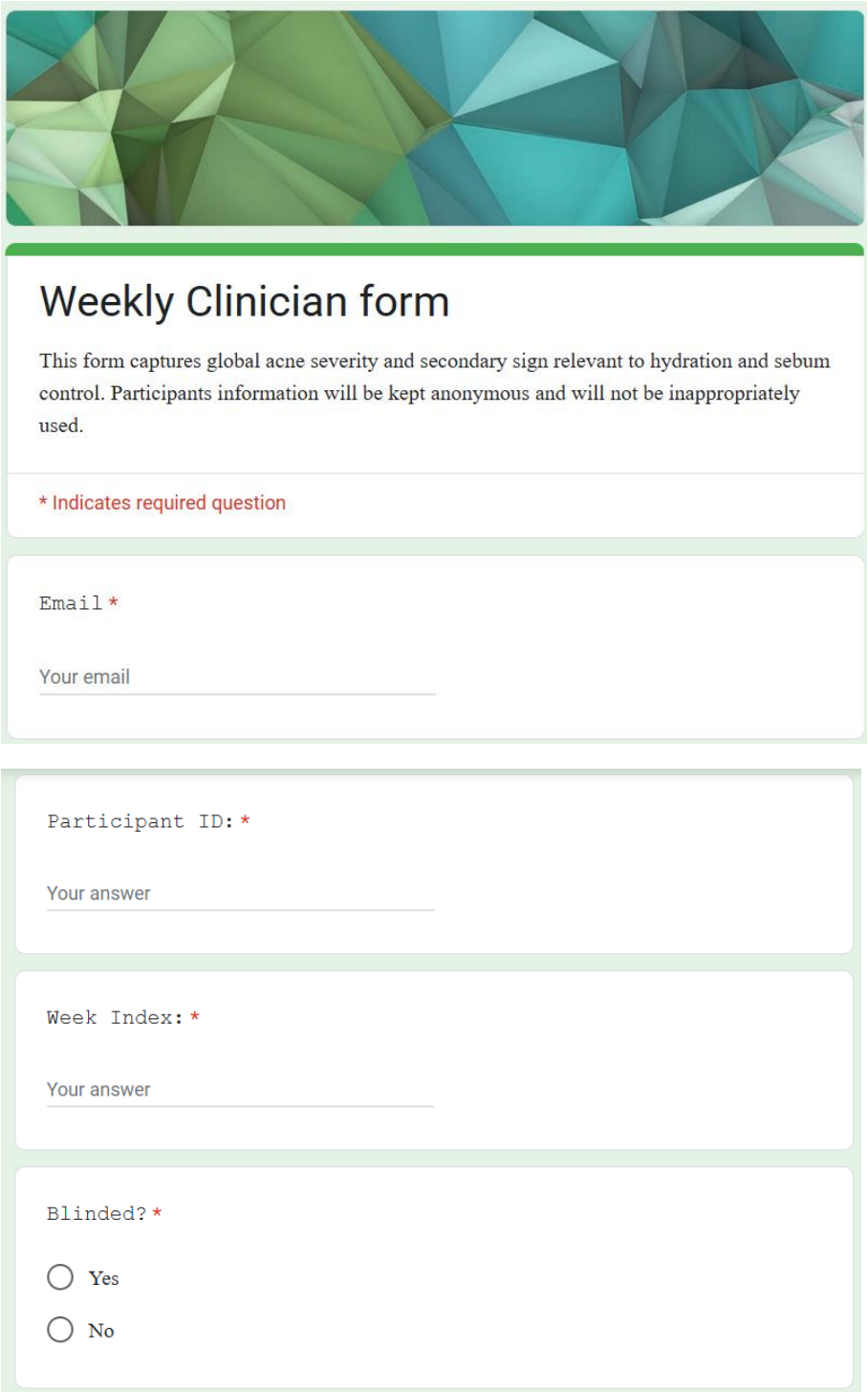
Figure 4

Standardized smartphone capture

Researcher capture record

- **Participant ID:**
- **Date/Time:**
- **Device model:**
- **Distance estimate:** cm
- **Pose error (yaw/pitch):** (degrees °)
- **Exposure clipping:** % dark / % bright
- **Blur score (Laplacian variance):**
- **Illumination notes:**
- **Accepted images:** Front | Left | Right
- **QC outcome:** Pass / Re-capture

Figure 5



Weekly Clinician form

This form captures global acne severity and secondary sign relevant to hydration and sebum control. Participants information will be kept anonymous and will not be inappropriately used.

* Indicates required question

Email *

Your email

Participant ID: *

Your answer

Week Index: *

Your answer

Blinded? *

- Yes
- No

Global acne severity: IGA 0-4 *

1 2 3 4

Erythema ? *

0 1 2 3

Pore Visibility?

0 1 2 3

Oiliness? *

0 1 2 3

Dryness? *

0 1 2 3

Notable lesions: (Count, location) *

Your answer

Safety Flags: Suspected adverse reaction? *

- ◯ Yes
- ◯ No

If Yes, describe

Your answer

Clinician ID: *

Your answer

Clinician Name: *

Your answer

Notes: *

Your answer

Submit Clear form